Polynomial Zeros and Turning-Point Constraints

Factoring to locate zeros, reading multiplicity from a factored form, bounding turning points by degree, and end behaviour of rational functions

Directions.

- Give exact zeros, and list every distinct real zero.
- A zero of **even** multiplicity touches the x-axis and turns back; a zero of **odd** multiplicity crosses it.
- A polynomial of degree n has at most n real zeros and at most $n-1$ turning points. Both bounds are ceilings, not counts.
- For end behaviour of a rational function, compare the degrees of the numerator and denominator.

1. Find all real zeros. $f(x) = x^2 - 7x + 10$

Answer: ________________

2. Let $g(x) = x^3 - 4x$.

(a) Factor $g(x)$ completely.

(b) State all real zeros of g.

Answer: ________________

3. Let $p(x) = (x-3)^2(x+1)$. List each distinct real zero with its multiplicity, and say whether the graph crosses or touches the x-axis there.

Answer: ____________________

4. A polynomial function h has degree 5.

(a) What is the greatest number of turning points the graph of h can have?

(b) What is the greatest number of distinct real zeros h can have?

Answer: ____________________

5. Find the end behaviour of the rational function below by evaluating the limit, and state the equation of the horizontal asymptote.

$$\lim_{x\to\infty} \frac{3x^2 - 5}{x^2 + 4}$$

Answer: ________________

6. Find all real zeros. One zero is a small integer — test the factors of the constant term first, then divide.

$$f(x) = x^3 - 2x^2 - 5x + 6$$

Answer: ________________

7. Find all distinct real zeros of $q(x) = x^4 - 6x^3 + 9x^2$ and give the multiplicity of each. How many times does the graph of q cross the x-axis?

Answer: __________________

8. The graph of a polynomial function f has exactly 4 turning points. What is the smallest possible degree of f? Explain in one sentence how the turning-point bound forces your answer.

Answer: __________________

9. Let $r(x) = x^4 - 5x^2 + 4$.

(a) Factor $r(x)$ completely. (Treat it as a quadratic in x^2.)

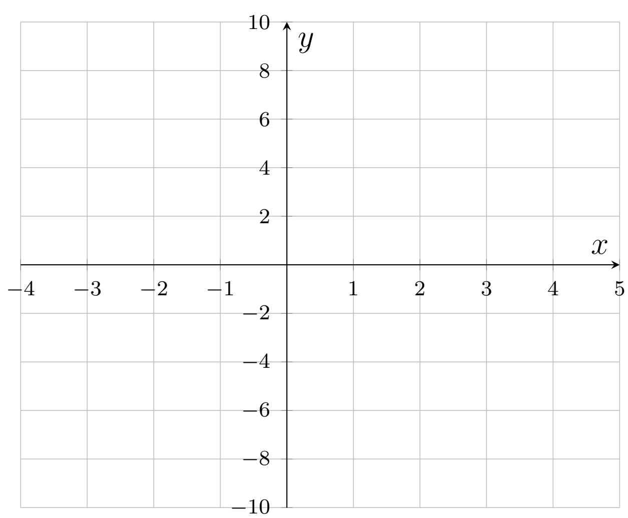
(b) State all real zeros, and say how many turning points the graph of r must therefore have.

Answer: ____________________

10. Let $s(x) = (x+2)(x-1)^2(x-3)$, a quartic with positive leading coefficient.

(a) List each distinct real zero with its multiplicity.

(b) Sketch a graph of s on the grid: mark the zeros, show a crossing or a touch at each one, and make the end behaviour correct. Exact y-values are not needed.



Answer: ____________________

11. Evaluate the limit and state the horizontal asymptote of the graph of $y = \dfrac{4x^2+3}{2x^2-x}$.

$$\lim_{x\to\infty} \frac{4x^2+3}{2x^2-x}$$

Answer: ________________

12. A student claims to have found a polynomial of degree 5 whose graph has 5 distinct real x-intercepts and only 2 turning points. Explain why no such polynomial exists. Your explanation should say what has to happen to the graph between two consecutive x-intercepts.